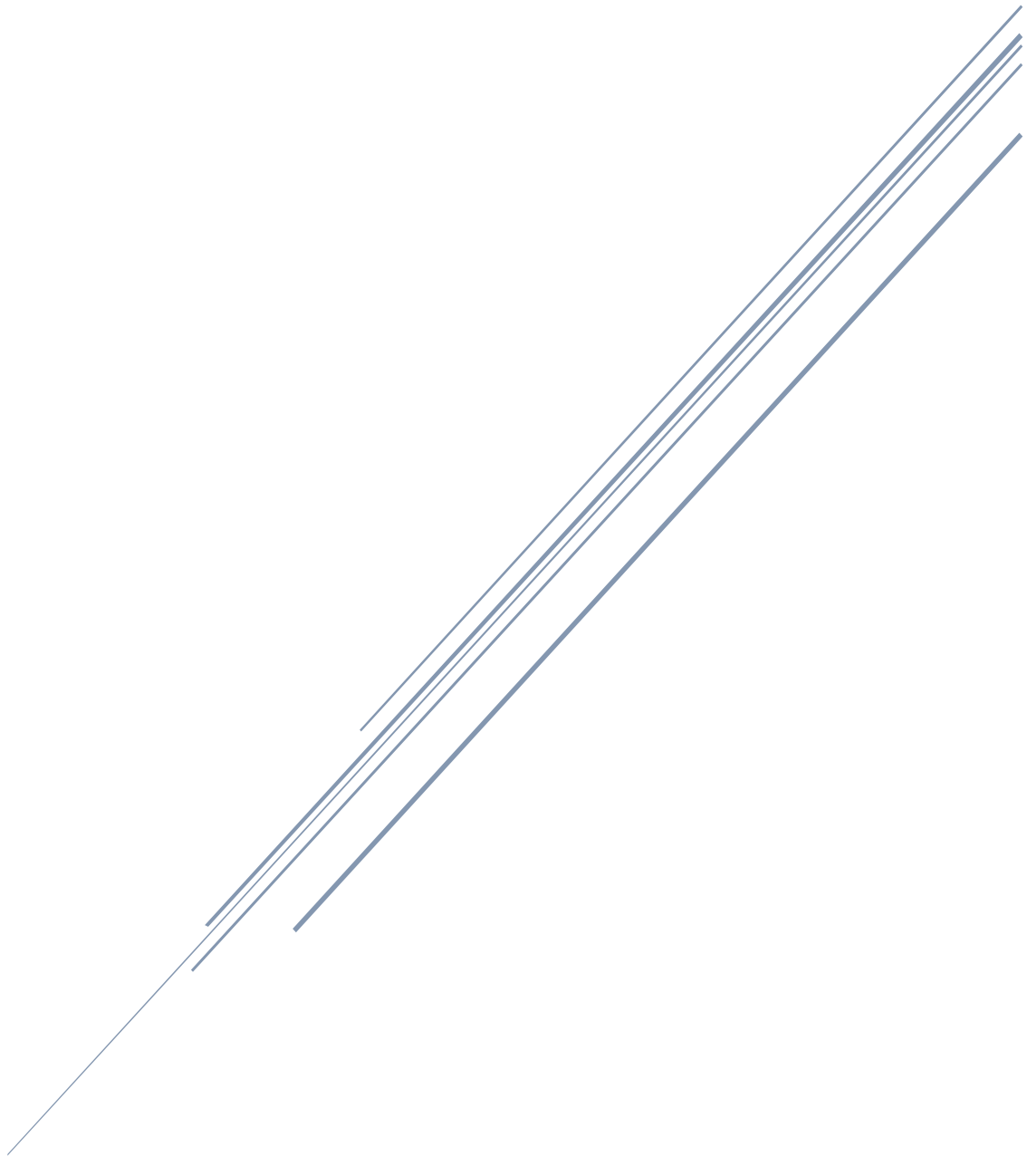


CHAPTER 5

Classification



Classification problems occur often, perhaps even more so than regression problems. Some examples include:

- A person arrives at the emergency room with a set of symptoms that could possibly be attributed to one of three medical conditions. Which of the three conditions does the individual have?
- An online banking service must be able to determine whether or not a transaction being performed on the site is fraudulent, on the basis of the user's IP address, past transaction history, and so forth.
- On the basis of DNA sequence data for a number of patients with and without a given disease, a biologist would like to figure out which DNA mutations are deleterious (disease-causing) and which are not.

Just as in the regression setting, in the classification setting we have a set of training observations $(x_1, y_1), \dots, (x_n, y_n)$ that we can use to build a classifier. We want our classifier to perform well not only on the training data, but also on test observations that were not used to train the classifier.

Example:

We will illustrate the concept of classification using the simulated Default data set. We are interested in predicting whether an individual will default on his or her credit card payment, on the basis of annual income and monthly credit card balance. The data set is displayed in Figure 5.1.

In the left-hand panel of Figure 5.1, we have plotted annual income and monthly credit card balance for a subset of 10,000 individuals. The individuals who defaulted in a given month are shown in orange, and those who did not in blue. (The overall default rate is about 3%, so we have plotted only a fraction of the individuals who did not default.) It appears that individuals who defaulted tended to have higher credit card balances than those who did not. In the center and right-hand panels of Figure 5.1, two pairs of boxplots are shown. The first shows the distribution of balance split by the binary default variable; the second is a similar plot for income. In this chapter, we learn how to build a model to predict default (Y) for any given value of balance (X_1) and income (X_2). Since Y is not quantitative, the simple linear regression model is not a good choice.

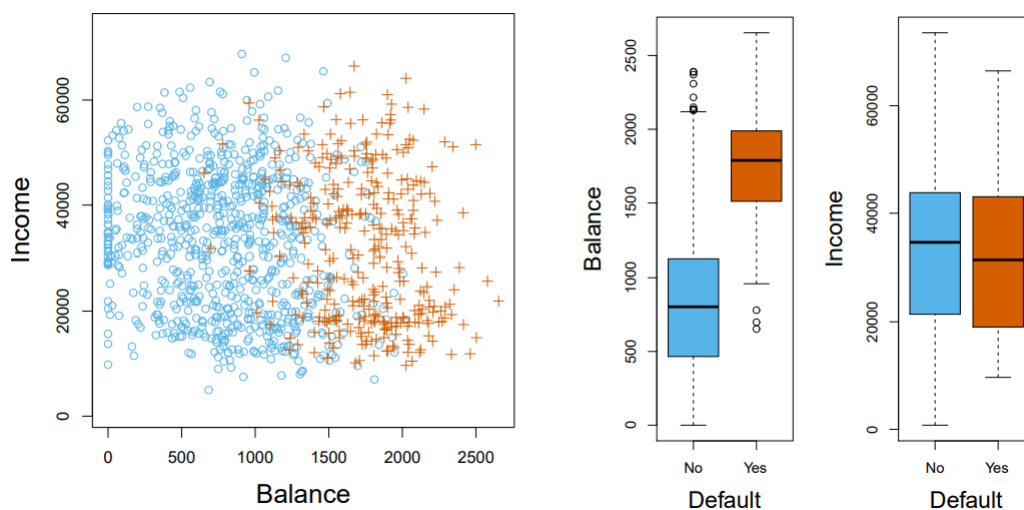


Figure 5.1 The Default data set. Left: The annual incomes and monthly credit card balances of a number of individuals. The individuals who defaulted on their credit card payments are shown in orange, and those who did not are shown in blue. Center: Boxplots of balance as a function of default status. Right: Boxplots of income as a function of default status.

5.1 Logistic Regression

Consider again the Default data set, where the response default falls into one of two categories, Yes or No. Rather than modeling this response Y directly, logistic regression models the probability that Y belongs to a particular category.

For the Default data, logistic regression models the probability of default. For example, the probability of default given balance can be written as

$$\Pr(\text{default} = \text{Yes} | \text{balance}).$$

The values of $\Pr(\text{default} = \text{Yes} | \text{balance})$, which we abbreviate $p(\text{balance})$, will range between 0 and 1. Then for any given value of balance, a prediction can be made for default. For example, one might predict $\text{default} = \text{Yes}$ for any individual for whom $p(\text{balance}) > 0.5$. Alternatively, if a company wishes to be conservative in predicting individuals who are at risk for default, then they may choose to use a lower threshold, such as $p(\text{balance}) > 0.1$.

5.1.1 The Logistic Model

How should we model the relationship between $p(X) = \Pr(Y=1|X)$ and X ? (For convenience we are using the generic 0/1 coding for the response.). If we considered using a linear regression model to represent these probabilities:

$$p(X) = \beta_0 + \beta_1 X.$$

These predictions are not correct, since of course the true probability of default, regardless of credit card balance, must fall between 0 and 1. To avoid this problem, we must model $p(X)$ using a function that gives outputs between 0 and 1 for all values of X . Many functions meet this description. In logistic regression, we use the logistic function.

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}.$$

$$\Pr(Y = 1/X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

$$\Pr(Y = 0/X) = 1 - \Pr(Y = 1/X) = \frac{1}{1 + e^{\beta_0 + \beta_1 X}}$$

Figure 5.2 illustrates the fit of the logistic regression model to the Default data. Notice that for low balances we now predict the probability of default as close to, but never below, zero. Likewise, for high balances we predict a default probability close to, but never above, one.

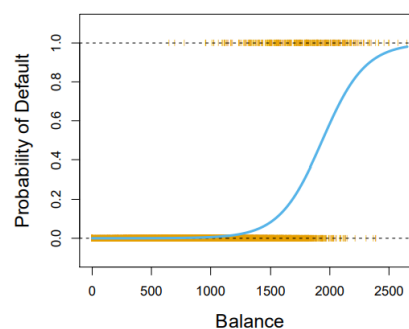


Figure 5.2 Classification using the Default data. Predicted probabilities of default using logistic regression. All probabilities lie between 0 and 1.

After a bit of manipulation of the previous equation, we find that

$$\frac{p(X)}{1 - p(X)} = e^{\beta_0 + \beta_1 X}.$$

The quantity $p(X)/[1-p(X)]$ is called the *odds*, and can take on any value between 0 and ∞ . Values of the odds close to 0 and ∞ indicate very low and very high probabilities of default, respectively.

By taking the logarithm of both sides, we arrive at

$$\log \left(\frac{p(X)}{1 - p(X)} \right) = \beta_0 + \beta_1 X.$$

The left-hand side is called the *log-odds* or *logit*. We see that the logistic regression model has a logit that is linear in X . The coefficients β tell us how much each feature increases or decreases the log-odds of being in class 1 rather than class 0.

5.1.2 Estimating the Regression Coefficients

The coefficients β_0 and β_1 are unknown, and must be estimated based on the available training data. In linear regression, we used the least squares approach to estimate the unknown linear regression coefficients. The more general method used in logistic regression is the *maximum likelihood*, since it has better statistical properties. The basic intuition behind using maximum likelihood to fit a logistic regression model is as follows: we try to find $\hat{\beta}_0$ and $\hat{\beta}_1$ such that plugging these estimates into the model for $p(X)$, yields a number close to one for all individuals who defaulted, and a number close to zero for all individuals who did not. This intuition can be formalized using a mathematical equation called a likelihood function:

$$\ell(\beta_0, \beta_1) = \prod_{i: y_i=1} p(x_i) \prod_{i': y_{i'}=0} (1 - p(x_{i'})).$$

The estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ are chosen to maximize this likelihood function. Maximum likelihood is a very general approach that is used to fit many of the non-linear models that we examine throughout this book. In the linear regression setting, the least squares approach is in fact a special case of maximum likelihood.

5.1.3 Making Predictions

Once the coefficients have been estimated, we can compute the probability of default for any given credit card balance. For example, if $\hat{\beta}_0 = -10.6513$ and $\hat{\beta}_1 = 0.0055$, we predict that the default probability for an individual with a balance of \$1,000 is

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 X}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 X}} = \frac{e^{-10.6513 + 0.0055 \times 1,000}}{1 + e^{-10.6513 + 0.0055 \times 1,000}} = 0.00576,$$

which is below 1%. In contrast, the predicted probability of default for an individual with a balance of \$2,000 is much higher, and equals 0.586 or 58.6%.

5.1.4 Multiple Logistic Regression

We now consider the problem of predicting a binary response using multiple predictors. By analogy with the extension from simple to multiple linear regression in Chapter 3:

$$\log \left(\frac{p(X)}{1 - p(X)} \right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p,$$

where $X = (X_1, \dots, X_p)$ are p predictors.

$$p(X) = \frac{e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}.$$

Just as simple logistic regression, we use the maximum likelihood method to estimate $\beta_0, \beta_1, \dots, \beta_p$.

5.1.5 Multinomial Logistic Regression

We sometimes wish to classify a response variable that has more than two classes. For example, we had three categories of medical condition in the emergency room: *stroke*, *drug overdose*, *epileptic seizure*. However, the logistic regression approach that we have seen in this section only allows for $K = 2$ classes for the response variable.

It turns out that it is possible to extend the two-class logistic regression approach to the setting of $K > 2$ classes. This extension is sometimes known as multinomial logistic regression. To do this, we first select a single class to serve as the baseline; without loss of generality, we select the K^{th} class for this role. Then we replace the model for logistic regression with the model

$$\Pr(Y = k|X = x) = \frac{e^{\beta_{k0} + \beta_{k1}x_1 + \cdots + \beta_{kp}x_p}}{1 + \sum_{l=1}^{K-1} e^{\beta_{l0} + \beta_{l1}x_1 + \cdots + \beta_{lp}x_p}}$$

for $k = 1, \dots, K-1$, and

$$\Pr(Y = K|X = x) = \frac{1}{1 + \sum_{l=1}^{K-1} e^{\beta_{l0} + \beta_{l1}x_1 + \cdots + \beta_{lp}x_p}}.$$

It is not hard to show that for $k = 1, \dots, K-1$,

$$\log \left(\frac{\Pr(Y = k|X = x)}{\Pr(Y = K|X = x)} \right) = \beta_{k0} + \beta_{k1}x_1 + \cdots + \beta_{kp}x_p.$$

An alternative coding for multinomial logistic regression, known as the softmax coding. The softmax coding is equivalent to the coding just described in the sense that the fitted values, log odds between any pair of classes, and other key model outputs will remain the same, regardless of coding. But the softmax coding is used extensively in some areas of the machine learning literature. In the softmax coding, rather than selecting a baseline class, we treat all K classes symmetrically, and assume that for $k = 1, \dots, K$,

$$\Pr(Y = k|X = x) = \frac{e^{\beta_{k0} + \beta_{k1}x_1 + \cdots + \beta_{kp}x_p}}{\sum_{l=1}^K e^{\beta_{l0} + \beta_{l1}x_1 + \cdots + \beta_{lp}x_p}}.$$

Thus, rather than estimating coefficients for $K - 1$ classes, we actually estimate coefficients for all K classes. The log odds ratio between the k^{th} and k'^{th} classes equals

$$\log \left(\frac{\Pr(Y = k|X = x)}{\Pr(Y = k'|X = x)} \right) = (\beta_{k0} - \beta_{k'0}) + (\beta_{k1} - \beta_{k'1})x_1 + \cdots + (\beta_{kp} - \beta_{k'p})x_p.$$